WANG ZHENG

(+86)13931805565 • neowang311@gmail.com • cheeseofneo.github.io

EDUCATION

The University of Tokyo – Tokyo, Japan Sep 2026 - July 2029
D.Eng., Dept. of Creative Informatics, Nakayama Lab
- *Thesis:* Physics-aware Perceptual Learning

Tsinghua University – Beijing, China Sep 2022 - July 2025
M.Eng., Dept. of Electrical Engineering 3.82/4.0 GPA
- *Thesis:* Research on a Fast Active Microscope Autofocus System Based on Off-Axis Illumination
- *Selective Course*: Advanced Signal Processing, Bayesian Statistics, Introduction to Neural Network Quantization

Tianjin University – Tianjin, China Sep 2018 - July 2022
B.Eng., Dept. of Electrical Engineering 3.72/4.0 GPA
- *Awards:* Membership of China Instrument and Control Society(2022),ICM (Interdisciplinary Contest in Modeling) Finalist(Top 1%, 2021), Merit Student(2021), Andon Trust Scholarship(2020), Liu Bao Scholarship(2019)
- *Thesis:* Research on Few-shot 3D Point Cloud Segmentation via Sample-Aware Data Augmentation
- *Selective Course*: Discrete Math, Theory of Computing, Deep Learning, Machine Learning, Pattern Recognition

HONORS

ICASSP, SIGGRAPH-Asia Reviewer, NVIDIA AI-Agent Group Member, Tsinghua College Scholarship

PUBLICATIONS

Wang Z., Ma R., Zeng C., Liu L., Li X., Wang X., & He B. A fast and precise autofocus method using linear array CCD. In Optical Metrology and Inspection for Industrial Applications XI (Vol.13241, pp.446-456).

Li Z., Su Y., Yang R., Xie C., **Wang Z.**, Xie Z., ... & Yang, H. (2025). Quantization meets reasoning: Exploring llm low-bit quantization degradation for mathematical reasoning. arXiv preprint arXiv:2501.03035.

Xiong H., Zhang J., **Wang Z.**, Pan T., & Hu Q. VividTalker: A modular framework for expressive 3D talking avatars with controllable gaze and blink. Accepted by ICASSP 2026.

Wang Z., Zhang R., Xiong H., Residual Persistence Guidance for next-scale image generation. arXiv preprint.

Xiong H.*, **Wang Z.***, et al., LiveEcho: Bridging Cyclic Visual Phase and Musical Progress for Live Photo-to-Music Generation. Under Review of AAAI.

Tang Z., Huang S., and **Wang Z.**, Retired-State Zero Exposure: Finite-Horizon Lifecycle Projection for Interactive Video Generation. Under Review of AAAI.

RESEARCH EXPERIENCE

Learning Collision-Aware Latent Dynamics for Physically Grounded Video Generation Feb 2026- Aug 2026
TeleAI, supervised by Liang Yuanzhi
- Designed a neural-operator framework that learns physical dynamics in a latent space to guide video diffusion models toward physically consistent generation. Introduced object-centered representations, latent operator-based dynamics prediction to address discontinuities and long-horizon rollout instability.

Residual Persistence Guidance for next-scale image generation Feb 2026- Aug 2026
UTokyo
- Developed an entropy-guided classifier-free guidance method for Visual Autoregressive image generation, establishing an information-theoretic framework that distinguishes epistemic from aleatoric uncertainty per spatial position; and demonstrated FID improvement on ImageNet-256.

Bridging Cyclic Visual Phase and Musical Progress for Live Photo-to-Music Generation. Dec 2025- April 2026
Under review of AAAI 27
- Propose the first system built for this gap of a temporal mismatch between a short looping visual clip with a longer, non-repeating soundtrack.
- Built a factorized conditioning framework (cyclic phase, monotonic progress, cover anchor) on a LoRA-adapted MusicGen backbone, trained on a self-curated dataset of 8,587 videocoveraudio triplets.

A Modular Framework For 3D Talking Avatars With Controllable Gaze And Blink Mar 2025- Sep 2025
Tsinghua, Published by ICASSP 2026

- Developed a modular pipeline integrating diffusion-based portrait synthesis, multilingual text-to-speech, 2D-to-3D lifting, and audio-driven animation, achieving 79% efficiency improvement over existing methods.
- Designed a physiologically-grounded gaze-and-blink enhancement module simulating human-like eye dynamics via coordinated saccadic movements, spontaneous blinking, and headeye compensation.

Exploring LLM Low-Bit Quantization Degradation for Mathematical Reasoning Dec 2024-Feb 2025
Research Assistant at **HKPolyU**, supervised by ***Prof.Yang Hongxia***,

- Proposed a novel framework for analyzing and mitigating quantization-induced reasoning errors in LLMs, which is mainly composed by step-aligned measurement suite, hierarchical (Conceptual, Method, Execution, Reasoning) error taxonomy and a compact Silver Bullet dataset enabling rapid recovery of math reasoning performance.
- Developed an automated chain-of-thought erroranalysis pipeline (judge ensemble and light human audit) that attains 97.2% labeling accuracy on 9,908 failure cases, enabling fine-grained, reproducible attribution by error type and first faulty step.

A fast and precise autofocus method using linear array CCD June 2023-Oct 2024
Tsinghua, supervised by ***Prof.Wang Xiaohao***

- Proposed a centroid extraction algorithm at sub-pixel level based on multiscale feature extraction, and fitted linear mapping using sliding finite impulse response filter. The final autofocus efficiency outperformed about 100 times than conventional gradient-based methods
- Developed a focus calibration algorithm based on weighted fusion to the results of several expert.

INTERNSHIP

TeleAI, Shanghai China: Research Intern,*Post Training Group* April 2026- Sep 2026
Research Topic: Physics-Consistent Video Generation

Baidu, Beijing China: Research Intern,*Team Apollo* July 2024- Oct 2024
Research Topic: Efficient Perceptual Model of Auto Vehicles.

Momenta, Shenzhen China: System Architecture Intern, *Team Middleware of System* March 2023-July 2023
Research Topic: Signal Simulation Platform for Perceptual Sensors of Auto Vehicles.

ADDITIONAL INFORMATION

Language: Chinese(Native), English(Fair), Japanese(Fluent)
Programming: Python, C, C++, MATLAB/Simulink, CUDA, Verilog
Certificate: IELTS 7.0, Duolinguo 130